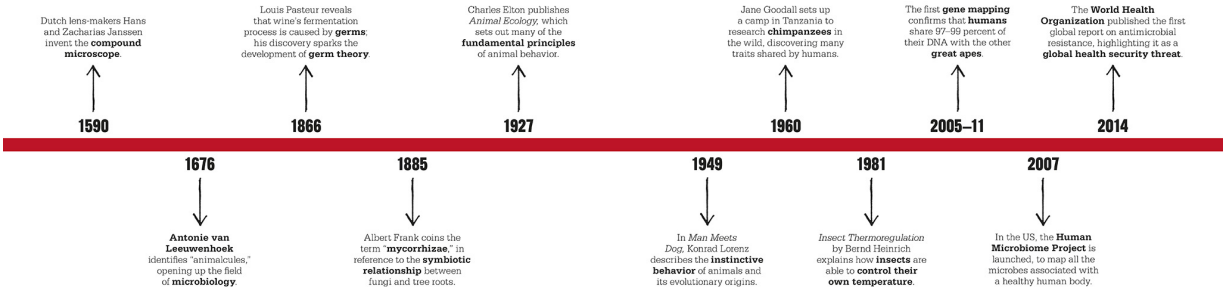
INTRODUCTION

Our understanding of the variety, behavior, and interaction of organisms has advanced considerably since Aristotle discovered that bee colonies have a queen and workers. Huge advances in technology, field observations, and laboratory experiments have increased our knowledge, and the modern study of animal behavior—ethology—continues to throw up surprises.



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Life under the microscope

Until the microscope was invented, no one knew that bacteria even existed, let alone what they did. Bacteria were first observed by Dutch microscopist Antonie van Leeuwenhoek in 1676, using an instrument he had built himself. He called these tiny organisms “animalcules,” but little was known about them for many years. In the 1860s, French chemist Louis Pasteur and German microbiologist Robert Koch developed the germ theory of disease, highlighting the harmful role played by bacteria. Subsequent research has also highlighted their positive roles: facilitating digestion; inhibiting the growth of other, pathogenetic bacteria; “fixing” or converting nitrogen into molecules that aid plant growth; and breaking down dead organic material, which releases nutrients for the food web.

Another discovery made possible by microscopy was of the mutualistic relationship between fungi and trees, published by German plant